

8.1 Implementation and Analysis of Decimation and Interpolation in Multirate Signal Processing Using MATLAB

```
clc;

clear;

close all;

% Original Sequence

n = 0:40;

x = sin(0.3*pi*n);

% Decimation by 2

y = x(1:2:end);

nd = 0:length(y)-1;

% Interpolation by 2

v = zeros(1,2*length(x));

v(1:2:end) = x;

ni = 0:length(v)-1;

% Figure 1: Original Sequence

figure;

stem(n,x,'filled');

grid on;

title('Original Sequence x[n]');

xlabel('n');

ylabel('Amplitude');

% Figure 2: Decimated Sequence
```

```
figure;  
stem(nd,y,'filled');  
grid on;  
title('Decimated Sequence by 2');  
xlabel('n');  
ylabel('Amplitude');
```

% Figure 3: Interpolated Sequence

```
figure;  
stem(ni,v,'filled');  
grid on;  
title('Interpolated Sequence by 2');  
xlabel('n');  
ylabel('Amplitude');
```

% Figure 4: Comparison

```
figure;  
stem(n,x,'b'); hold on;  
stem(0:2:length(v)-2,v(1:2:end),'r');  
legend('Original x[n]', 'Interpolated Samples');  
grid on;  
title('Comparison of Sequences');  
xlabel('n');  
ylabel('Amplitude');
```



